

ASTANA IT UNIVERSITY
FACULTY OF MANAGEMENT
AI BUSINESS

Final Exam | Report

AI-Driven Smart Retail Supply Chain Intelligence System

Discipline: AI for Logistics and Supply Chain
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1. Introduction

This project focuses on building an AI-driven retail supply chain intelligence system using a retail sales dataset from Kaggle. The main purpose of the project is to analyze sales and supply chain patterns, forecast future demand, predict delivery delay risk, optimize inventory, and evaluate supplier and transportation performance.

Retail supply chains require accurate demand planning, stable inventory management, and efficient logistics operations. If demand is not predicted correctly, a company may face stockouts, overstock, high transportation costs, or delivery delays. Because of this, data analytics and machine learning can help businesses make better and faster supply chain decisions.

In this project, Python was used for data cleaning, exploratory data analysis, machine learning, forecasting, clustering, and risk analysis. Power BI will be used for visual reporting and dashboard development.

2. Dataset Description

The project uses three connected datasets from Kaggle: sales data, features data, and stores data.

The sales dataset contains weekly sales information by store, department, date, and holiday status. The main target variable in this file is `Weekly_Sales`, which represents demand.

The features dataset contains external and economic factors such as temperature, fuel price, markdown promotions, CPI, unemployment, and holiday status. These variables are useful because retail demand and logistics performance can be affected by promotions, economic conditions, holidays, and fuel prices.

The stores dataset contains store-level information, including store type and store size. Store type and size are important because different store formats may have different demand patterns and operational needs.

The datasets were merged using `Store`, `Date`, and `IsHoliday`. After merging, the final dataset included sales information, store characteristics, economic indicators, promotional markdowns, and time-based features.

3. Methodology

The project was divided into six main analytical modules:

- Data Cleaning
- Exploratory Data Analysis
- Demand Forecasting
- Delivery Delay Prediction
- Inventory Optimization
- Supplier and Transportation Analytics

Each module supports a different part of supply chain decision-making. Together, these modules create a full AI-driven retail supply chain intelligence system.

4. Task 1: Data Cleaning

In the data cleaning stage, the three datasets were loaded and checked for missing values, duplicates, data types, and general structure.

The Date column was converted into datetime format. The sales dataset was merged with the features dataset using Store, Date, and IsHoliday. After that, the merged dataset was connected with the stores dataset using Store.

Most missing values appeared in the markdown columns. These missing values were replaced with 0 because missing markdown values usually mean that no promotion was applied. Missing values in CPI and Unemployment were filled using the median value because the median is more stable and less affected by extreme values.

Additional features were created to improve analysis and modeling. These included Year, Month, Week, Quarter, Total_MarkDown, Sales_per_Size, and IsHoliday_Int.

As a result, the dataset became clean, complete, and ready for exploratory data analysis and machine learning.

5. Task 2: Exploratory Data Analysis

Exploratory data analysis was used to understand the main sales patterns in the dataset.

The total weekly sales trend showed clear seasonal peaks. The strongest increases appeared around holiday periods, especially near the end of the year. This means that the business should prepare additional inventory and logistics capacity before high-demand weeks.

The yearly sales analysis showed that 2011 had the highest total sales. Sales in 2012 were lower mainly because the dataset ends in October 2012 and does not include the full year.

The monthly sales analysis showed that demand is not equally distributed across the year. Some months had stronger sales activity, which confirms the presence of seasonality in retail demand.

Holiday weeks had higher average weekly sales than non-holiday weeks. This shows that holidays are important demand drivers and should be included in forecasting and inventory planning.

Store Type A generated the highest total sales, followed by Type B and Type C. This means that Type A stores are the most important for revenue and should receive stronger inventory and supply chain support.

The top-performing stores included Store 20, Store 4, Store 14, Store 13, and Store 2. The top-performing departments included 92, 95, 38, 72, and 90. These stores and departments should be prioritized in forecasting, replenishment, and inventory planning.

The correlation matrix showed that store size has a positive relationship with weekly sales. Markdown variables had weaker direct correlation with sales, which means that promotions may affect demand but not always in a simple linear way.

The outlier analysis showed many high weekly sales values. These outliers were not removed because they may represent real business events such as holidays, promotions, seasonal peaks, or high-performing departments.

6. Task 3: Demand Forecasting

The demand forecasting module was developed to predict future sales and support inventory planning. The target variable was Weekly_Sales, because it represents retail demand.

Four forecasting methods were used: Linear Regression, Random Forest Regressor, ARIMA, and Prophet.

Linear Regression was used as a baseline model. Its performance was weak, with an R2 score of about 0.089. This means that a simple linear model cannot explain weekly sales well. Retail demand is more complex and depends on non-linear relationships between store, department, season, holiday, promotion, and economic variables.

Random Forest Regressor performed much better, with an R2 score of about 0.973. This means that the model explained most of the variation in weekly sales and produced more accurate predictions. The Actual vs Predicted graph also showed that Random Forest predictions followed real sales values closely.

The feature importance results showed that Department was the most important variable for demand forecasting. Store size, store number, week, and CPI also influenced sales. This means that demand planning should be done by department, store, and time period instead of using only total company-level sales.

ARIMA was used as a traditional time-series forecasting method. It predicted future monthly sales based on historical sales trends. ARIMA worked as a simple time-series benchmark and showed a stable forecast for the next months.

Prophet was also used for monthly forecasting. It was useful because it provided forecasted values together with lower and upper forecast ranges. This helps the business understand possible uncertainty in future demand.

Based on the Prophet forecast, future inventory need levels were created. High forecasted demand was linked with high inventory need, while lower forecasted demand was linked with low inventory need. This connects demand forecasting with inventory planning.

Overall, Random Forest was the strongest model for weekly sales prediction, while Prophet was useful for monthly business forecasting and future inventory planning.

7. Task 4: Delivery Delay Prediction

The original retail sales dataset does not contain direct logistics variables such as shipping mode, warehouse, region, distance, traffic, or order priority. However, the project required a delivery delay prediction module. Therefore, a proxy logistics layer was created using the available retail data.

Store numbers were grouped into regions. Each region was connected to a warehouse. Store type was used as a proxy for shipping mode. Weekly sales level was used to create order priority. Traffic index was estimated using fuel price, holiday weeks, and sales pressure.

The delivery delay target was created using the logistic probability formula:

$$P(Y=1) = 1 / (1 + e^{-(\beta_0 + \beta_1 X)})$$

The formula included logistics risk factors such as long distance, high traffic, high priority orders, holiday weeks, and high fuel price.

Three classification models were used: Logistic Regression, Decision Tree Classifier, and Random Forest Classifier.

Logistic Regression achieved about 94% accuracy. It was useful because it directly follows the logic of delay probability and is easy to explain.

Decision Tree and Random Forest Classifier achieved almost perfect accuracy. This result is expected because the delay target was generated from clear logistics rules, and the models could easily learn those patterns.

The feature importance results showed that Traffic Index was the strongest driver of delivery delay prediction. Fuel price, holiday weeks, distance, and sales pressure were also important. The high-risk route analysis showed that critical priority orders in Central, South, East, and West regions had higher delay probabilities. These routes should be monitored carefully, especially during holiday periods and high-demand weeks. Overall, this module provides delay probabilities, high-risk route identification, and delivery performance insights.

8. Task 5: Inventory Optimization

The inventory optimization module used ABC Analysis, K-Means Clustering, overstock and understock detection, and reorder point recommendations.

The dataset does not contain direct product stock quantities. Therefore, departments were used as product categories, and weekly sales were used as a demand proxy.

ABC Analysis classified departments based on their contribution to total sales. Category A included 28 departments and generated around 79.33% of total sales. Category B included 20 departments and contributed around 15.53% of sales. Category C included 33 departments and contributed around 5.14% of sales.

This result confirms the classic ABC logic: a smaller group of high-value departments creates most of the revenue. Category A departments should be monitored more closely because stockouts in these departments can strongly affect total sales.

K-Means Clustering was applied using total sales, average weekly sales, demand volatility, store coverage, and average markdown. The goal of K-Means was to group departments with similar demand and inventory behavior by minimizing the distance between each department and its cluster center.

The stock status analysis identified 34 departments as Normal Stock, 36 departments as Overstock Risk, and 11 departments as Understock Risk. Overstock risk means that departments may have lower demand but higher markdown activity. Understock risk means that departments have high demand and high volatility, so they may need faster replenishment.

Reorder points were calculated using average weekly demand, estimated lead time, and safety stock. The highest reorder point departments included 92, 95, 38, 72, and 90. These departments should be monitored weekly and should maintain enough safety stock to avoid stockouts.

Overall, this module provides data-driven inventory categories, cluster groups, stock status, reorder points, and reorder recommendations.

9. Task 6: Supplier and Transportation Analytics

The supplier and transportation analytics module was completed using a proxy supply chain network.

The original retail dataset does not contain direct supplier names, real routes, transportation cost, or lead time. Therefore, warehouses were treated as suppliers or distribution partners. Regions were created from store groups, and shipping modes were created from store types. Transportation performance was estimated using distance, fuel price, traffic index, delivery delay, and weekly sales.

Supplier reliability was measured using delay rate, lead time, transportation cost, route efficiency, fuel price, and traffic index. A reliability score was created to rank suppliers.

The supplier ranking showed that Warehouse_E had the highest reliability score. It had a lower delay rate, shorter lead time, and stronger route efficiency. Warehouse_S had the lowest reliability score, mainly because it had a higher delay rate and longer lead time.

Transportation cost analysis showed that West and South routes were the most expensive, especially when Same Day or Second Class shipping was used. This was mainly caused by longer average distance and higher delivery pressure.

Route efficiency was measured as sales generated per unit of transportation cost. Routes with higher route efficiency are better for the supply chain because they generate more business value with lower logistics pressure.

Fuel trend analysis showed that fuel price strongly affects transportation cost. The scatter plot confirmed that higher fuel prices are connected with higher average transportation cost.

The bottleneck analysis identified one critical bottleneck and four moderate bottlenecks. The most critical route was West / Warehouse_W / Second Class because it had high delay rate, high transportation cost, and low route efficiency.

The supply chain risk analysis identified three high-risk routes. These were mainly West and South routes with Same Day or Second Class shipping. These routes should be monitored carefully, especially during holiday periods and high-demand weeks.

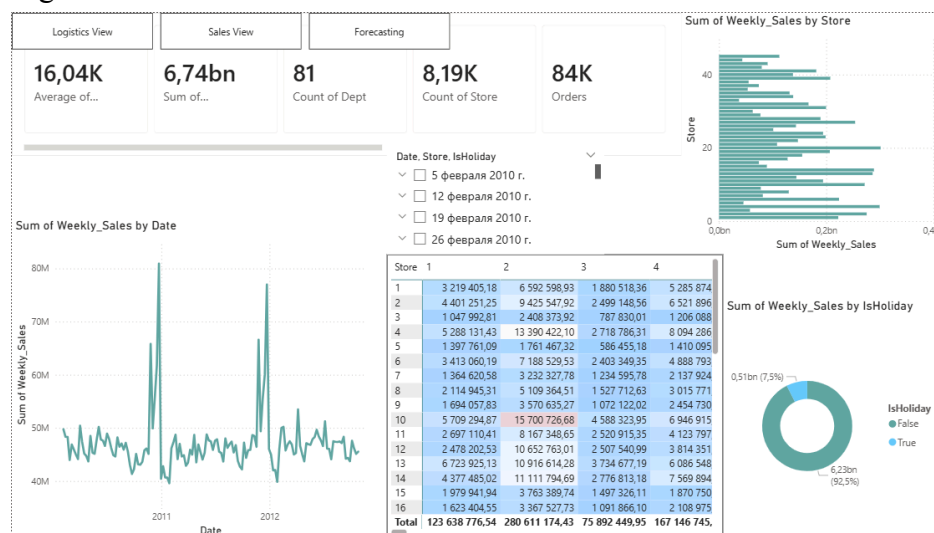
Overall, this module provides supplier rankings, cost optimization insights, bottleneck identification, and supply chain risk analysis.

10. Power BI Dashboard Description

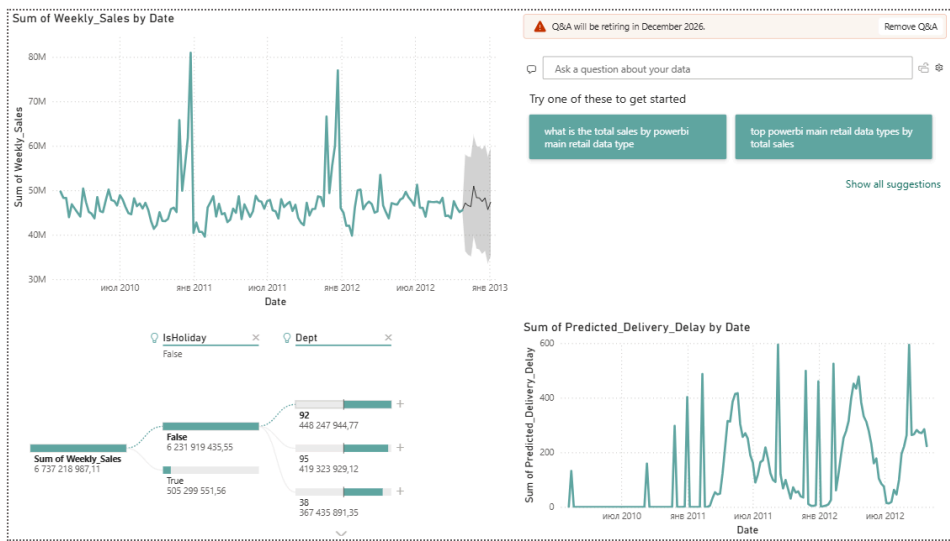
This section will be completed after the Power BI dashboard is created.

Dashboard Pages

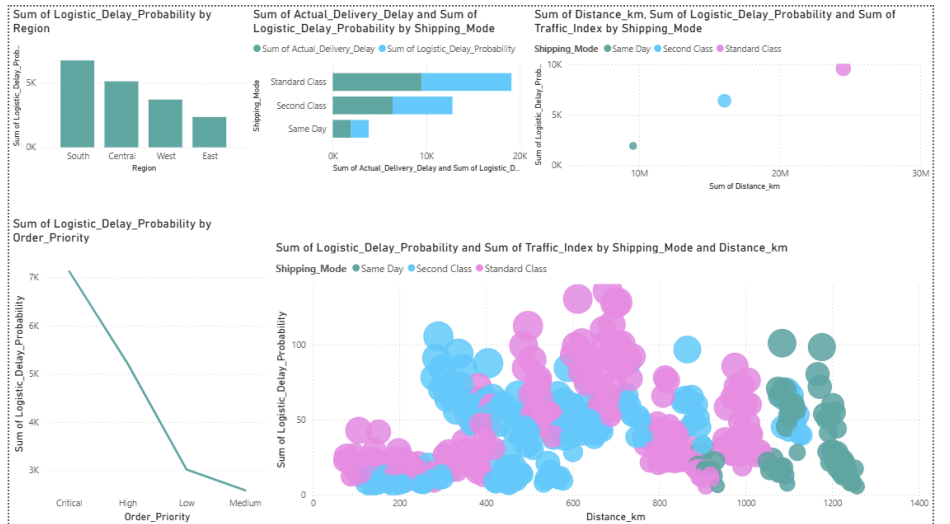
Page 1: Executive Dashboard



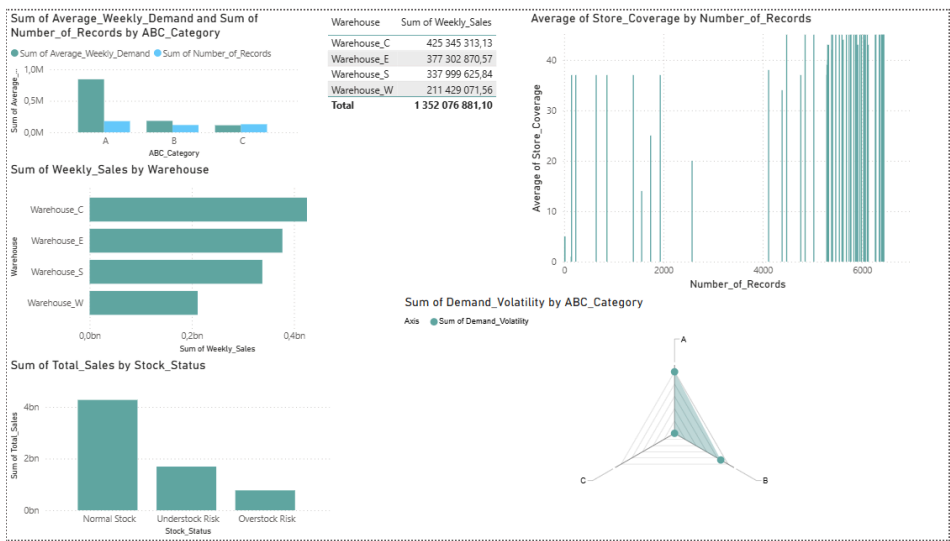
Page 2: Demand Forecasting



Page 3: Delivery Delay and Logistics Risk



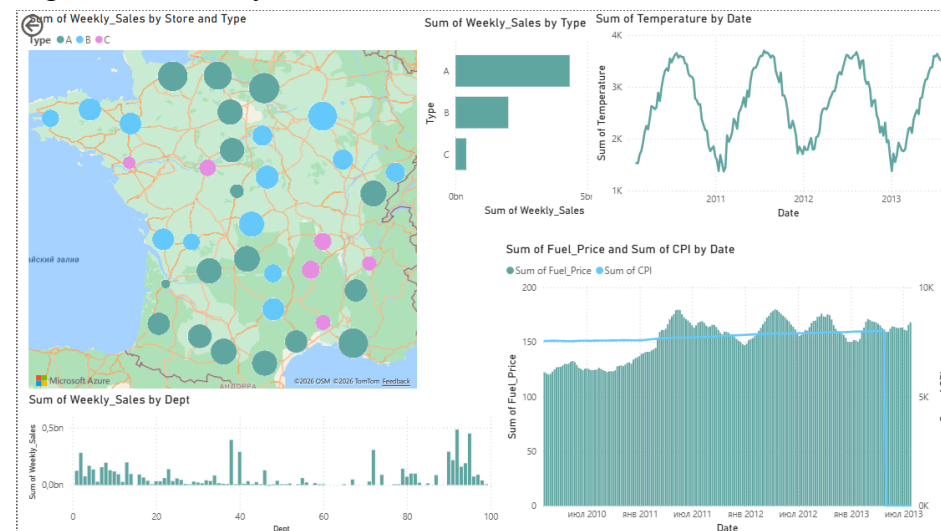
Page 4: Inventory Intelligence



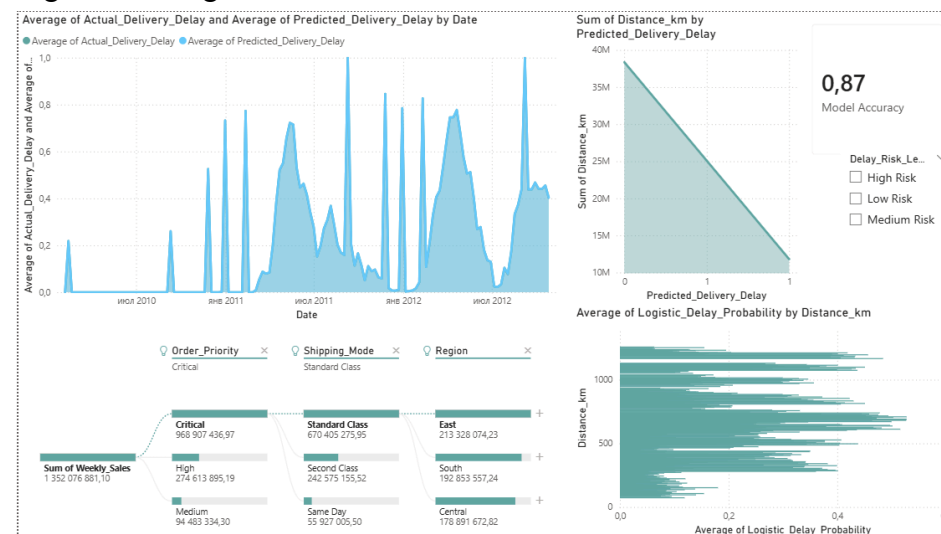
Page 5: Supplier and Transportation Analytics



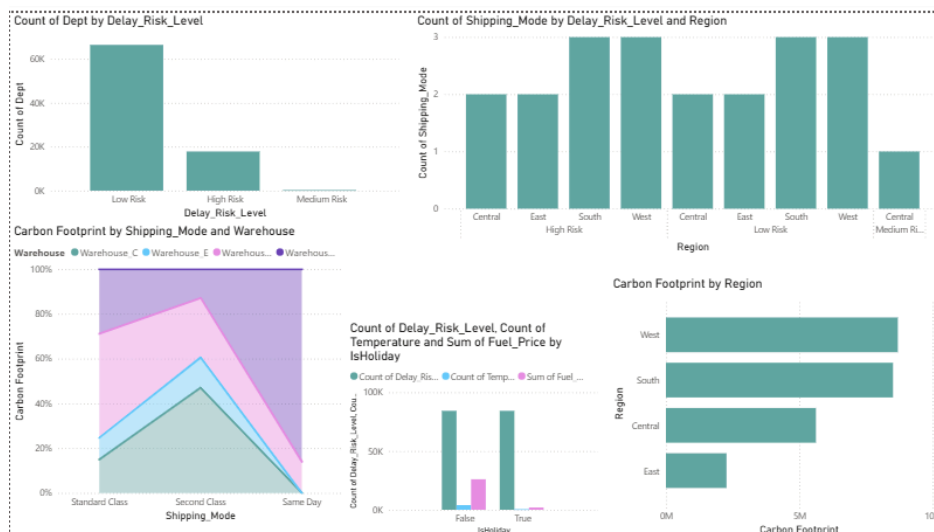
Page 6: Sales Analysis



Page 7: AI insights



Page 8: Risk



Page 9: Strategic Recommendations

Strategic Recommendations

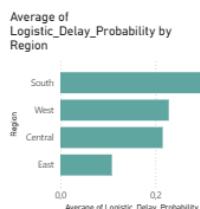
To improve performance, the company should focus on reducing delays in high **Traffic_Index** regions and optimizing long-distance routes (**Distance_km**) to lower costs and improve efficiency.

Inventory management must be strengthened by increasing stock levels in stores with low **Store_Coverage** to avoid stockouts.

High-risk deliveries (**Delay_Risk_Level**) should be prioritized to improve reliability.

AI models should be used more actively for demand forecasting and logistics planning to support better decision-making.

Finally, optimizing transportation and shipping modes will help reduce carbon emissions and support sustainability goals.



Dashboard Screenshots

11. Key Business Insights

The first key insight is that sales are seasonal and increase during holiday periods. This means the company should prepare additional inventory and logistics capacity before high-demand weeks.

The second key insight is that Random Forest is the strongest model for weekly demand prediction. This means retail demand is non-linear and cannot be fully explained by simple linear relationships.

The third key insight is that Category A departments generate most of the revenue. These departments should receive priority in replenishment planning and safety stock management.

The fourth key insight is that delivery delay risk increases during high traffic, high fuel price, holiday weeks, long-distance routes, and critical priority orders.

The fifth key insight is that West and South routes have higher transportation cost and supply chain risk. These routes need more attention in cost optimization and logistics planning.

12. Recommendations

The company should use machine learning forecasting for weekly demand planning, especially Random Forest, because it gives strong prediction accuracy.

The company should monitor Category A departments weekly and maintain enough safety stock for high-demand and high-volatility departments.

High-risk delivery routes should be monitored before holiday periods. The company should prepare earlier shipments and additional transport capacity for critical orders.

Warehouse_S and Warehouse_W routes should be reviewed because they show higher risk or lower reliability compared with other warehouses.

Fuel price should be included in logistics budgeting because transportation cost increases when fuel prices rise.

Power BI dashboards should be used by managers to track sales, forecasted demand, inventory risk, delivery delays, supplier reliability, and transportation bottlenecks in one system.

13. Conclusion

This project developed an AI-driven smart retail supply chain intelligence system using a retail sales dataset from Kaggle. The system combines data cleaning, exploratory analysis, demand forecasting, delivery delay prediction, inventory optimization, and supplier and transportation analytics.

The results show that machine learning and business intelligence can help improve supply chain planning. The project supports better decisions in demand forecasting, inventory management, logistics risk control, supplier ranking, and transportation cost optimization.

Although some logistics variables were not directly available in the dataset, proxy features were created to complete the required supply chain modules. This made it possible to build a full analytical pipeline and prepare outputs for Power BI reporting.

14. References

Kaggle Dataset: Retail Dataset by Manjeet Singh

Python Libraries: pandas, numpy, matplotlib, seaborn, scikit-learn, statsmodels, Prophet

Power BI: Microsoft Power BI for dashboard visualization